

Majorana Modes in the Machine

Week 6: From One Measurement to a Phase Diagram

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Week 6 Objective

Goal

Turn the Week 5 proof-of-principle into a diagnostic: sweep μ and ask whether a gate-measurable non-local correlator reconstructs the topological phase boundary.

- Week 5 measured one sweet-spot example at $t = \Delta$, $\mu = 0$.
- Week 6 varies μ across trivial, critical, and topological regimes.
- The target observable remains parity-preserving and non-local.
- The emphasis now shifts from solving the model to diagnosing phases operationally.

The Observable We Keep

$$O_{\text{edge}} = X_0 Z_1 \cdots Z_{L-2} X_{L-1}$$

- Single-edge operators flip fermion parity, so their expectation vanishes in a parity eigenstate.
- Unlike Ising-type order parameters, topology cannot generally be detected locally.
- The Jordan–Wigner mapping converts an edge-to-edge Majorana correlator into a non-local Pauli string.
- The two-edge string preserves parity and probes coherent long-range edge correlations.
- For $L = 4$:

$$O_{\text{edge}} = X_0 Z_1 Z_2 X_3.$$

Ising vs Kitaev: What is the Order?

Ising (Symmetry Breaking)

- Order parameter: $\langle \sigma^z \rangle$
- Phase defined by symmetry breaking
- Local correlations detect the phase
- Excitations are domain walls

Kitaev Chain (Topology)

- No local order parameter
- Phase defined globally
- Edge modes encode information
- Excitations are fermionic quasiparticles

Key Difference

Ising phases are local and symmetry-based. Kitaev phases are global and information-based.

Why Local Probes Fail in the Kitaev Chain

Ising intuition works because

- Local magnetization captures the phase
- Order parameter is directly measurable

But in Kitaev systems

- All bulk phases look locally similar (gapped)
- No local observable distinguishes topology
- Edge physics is invisible to local measurements

Conclusion

A non-local observable is required to access topology experimentally.

Symmetry Breaking vs Topology

Ising Hamiltonian

$$H = J \sum_i Z_i Z_{i+1} - h \sum_i X_i$$

Ising Chain

Ordered



$$\langle \sigma^z \rangle \neq 0$$

Disordered



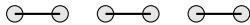
$$\langle \sigma^z \rangle = 0$$

Kitaev (JW qubit form)

$$H = \frac{t + \Delta}{2} \sum_j Y_j Y_{j+1} - \frac{\mu}{2} \sum_j Z_j$$

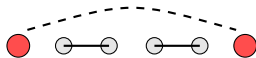
Kitaev Chain

Trivial



$$|\mu| > 2t$$

Topological



$$|\mu| < 2t$$

Measurement Protocol

Circuit-level measurement

- Prepare the ground state using the VQE ansatz.
- Rotate qubits with X factors into the Z basis using Hadamards.
- Measure all qubits and multiply bit parities to estimate $\langle O_{\text{edge}} \rangle$.
- The protocol converts a topological property into experimentally accessible bitstring statistics.
- The observable is hardware-compatible because it is a Pauli string.

Shot estimator ($N = 4096$)

Each shot returns an eigenvalue $m_s = \pm 1$.

$$\langle O \rangle \approx \frac{1}{N} \sum_{s=1}^N m_s, \quad \sigma \approx \sqrt{\frac{1 - \langle O \rangle^2}{N}}$$

At $N = 4096$, $\sigma \lesssim 0.016$ in the topological window. Error bars are omitted from sweep plots for clarity.

Expected Physical Behavior

Topological

- Edge modes separated
- Finite string order
- Coherent boundaries

$$|\mu| < 2t$$

Critical

- Bulk gap closes
- Correlation diverges
- Edge delocalization

$$|\mu| = 2t$$

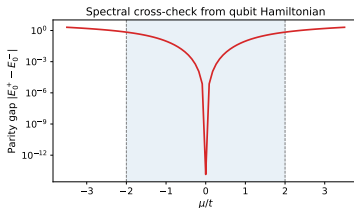
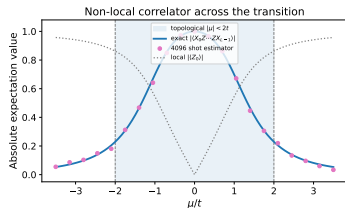
Trivial

- Edge modes hybridize
- No long-range order
- String collapses

$$|\mu| > 2t$$

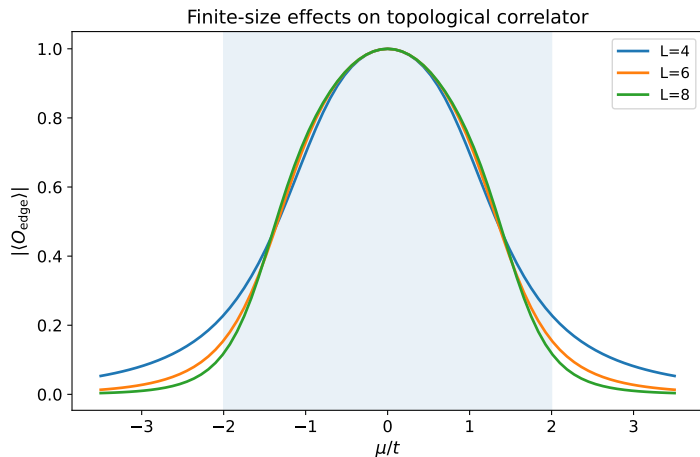
Numerical Phase Sweep

Week 6: Measuring topology by sweeping μ ($L = 4$, $t = \Delta = 1$)



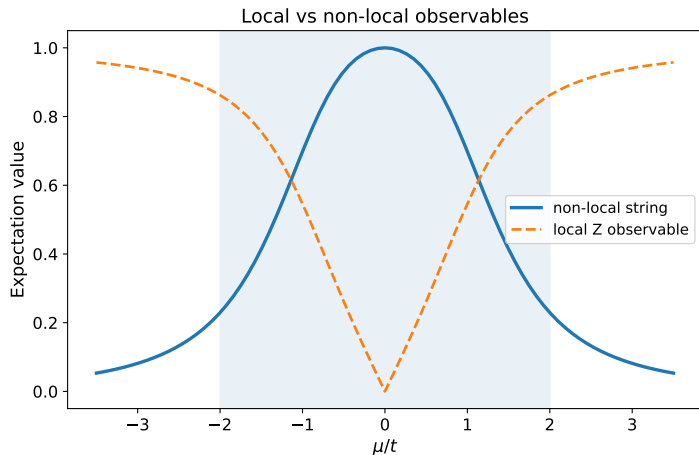
- Parameters fixed:
 $t = \Delta = 1$; sweep is 1D in μ
- Topological window:
 $|\mu| < 2t$
- String correlator detects edge coherence
- Sharp decay outside phase
- Shot estimator follows same trend
- Parity gap matches transition point

Finite-Size Scaling



- Small L : smooth crossover
- Larger L : sharper boundary
- $L = 4, 6, 8$ remain far from the thermodynamic limit; the sharpening *trend* is the key result
- Approaches thermodynamic limit
- Edge coherence becomes well-defined

Local vs Non-Local Observables



- Local Z_0 : weak phase signal
- Non-local string: sharp diagnostic
- Topology is not locally visible
- Edge physics is inherently global

Implementation Step

```
for mu in mu_values:
    H = kitaev_qubit_hamiltonian(L, t, mu, delta)
    psi = even_parity_ground_state(H)    # exact diagonalization, NOT VQE
    value = <psi| X0 Z1 ... Z(L-2) X(L-1) |psi>
    shot_value = sample_pm_one(value, shots=4096)
```

- **Note:** `even_parity_ground_state` uses exact diagonalization – not VQE. The pseudocode masks this gap; a real VQE loop is the next milestone.
- Exact diagonalization is valid only for small L ; it serves as a validation benchmark, not a scalable method.
- The measured quantity is the same Pauli string a circuit would estimate from bitstrings.
- The same workflow can be executed on quantum hardware using repeated circuit measurements.
- The next code revision should plug this sweep into the VQE runner directly.

Main Conclusions

- Topological phases require non-local observables.
- A measurable Pauli-string correlator reconstructs the Kitaev phase boundary.
- The diagnostic survives finite-size effects and statistical sampling.
- Quantum measurements can therefore operationally detect topology.

Conceptual Message

The transition is not characterized by symmetry breaking, but by the emergence of long-range topological coherence.

Next: Block 4

- Add a real VQE loop for every μ point instead of ED-prepared validation states.
- Attach depolarizing, readout, and relaxation noise models to the measurement circuits.
- Compare ideal, noisy, and mitigated string-order curves.
- Study robustness of non-local topology diagnostics under realistic imperfections.
- Investigate how much noise the phase boundary tolerates before the signal becomes unreliable.
- Extend sweep to a 2D (t, μ) phase diagram.